Combined Cell

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1 Combined Cell Overview

> The Combined Cell feature allows two or more sector carriers to be operated as a single cell.

Feature Name	Combined Cell
Feature Identity	FAJ 121 3025
Value Package Name	Combined Cell
Value Package Identity	FAJ 801 0407
Node Type	Baseband Radio Node
Access Type	LTE
Licensing	License-controlled feature. One license is required for each radio.

Summary

The Combined Cell feature extends the indoor and outdoor coverage of a single cell.
This feature provides the following benefits:

- Reduces coverage holes by allowing multiple coverage areas within the same cell.
- Enhances coverage quality by selecting sector carriers with the best coverage quality.
- Eliminates the need for inter-cell handover within a combined cell.
- Supports uplink macrodiversity by using multiple antennas within the same cell.

2 Dependencies of Combined Cell

> The feature has dependencies with other features that have an impact on carrier aggregation, downlink, and uplink capabilities.

Feature Dependencies

Table 1 Feature Dependencies

Feature Name	Feature Identity	Access Type	Relationship	Description
4x2 Quad Antenna Downlink Performance Package	FAJ 121 3041	LTE	Related	The 4x2 Quad Antenna Downlink Performance Package feature is needed when the Combined Cell feature is used with four TX and four RX antennas, to support up to two layers spatial multiplexing (TM3 and TM4).
4x4 Full Interference Rejection Combining	FAJ 121 4901	LTE	Related	If this feature is used along with Combined Cell, then the Combined Cell feature uses full IRC in groups of 4RX for scenarios with four receive antennas.
4x4 Quad Antenna Downlink Performance Package	FAJ 121 3076	LTE	Related	The 4x4 Quad Antenna Downlink Performance Package feature is needed when the Combined Cell feature is used with four Tx and four Rx antennas, to support up to four layers spatial multiplexing (TM3 and TM4).
Carrier Aggregation	FAJ 121 3046	LTE	Related	The downlink sector carrier selection algorithm of the Combined Cell feature is not applied to the secondary serving cell in a carrier aggregation configuration.
Category M Access	FAJ 121 4658	MI	Related	Combined Cell can coexist with the Category M Access feature but there is potential performance impact on non-static Category M1 UEs, because PUCCH CQI is not used for Category M. There is no sector carrier switch after the initial sector carrier is determined at the random access procedure. This can cause connection interruptions for Category M1 UEs that are in connected mode for long periods and move to a new sector carrier.
Data-Aware Carrier Aggregation Configuration	FAJ 121 5116	LTE	Related	A combined cell with more than two sector carriers cannot be configured as an uplink SCell.
High Speed UE	FAJ 121 2054	LTE	Related	When the High Speed UE feature is used together with the Combined Cell feature, uplink sector combining must be disabled by setting `ENodeBFunction.combCellSectorSelectThreshRx` to zero.
Intelligent Power Emission Control	FAJ 121 5224	LTE	Related	The Intelligent Power Emission Control feature and the Combined Cell feature can be used together only in a non-AAS deployment. The features are not supported together in an AAS deployment.
LTE PIM Detection	FAJ 121 5436	LTE	Related	When the Combined Cell feature is active, one cell maps to multiple sector carriers. This makes analyzing relationships between cells and sector carriers more ambiguous, which reduces the resolution and usability of data provided by the LTE PIM Detection feature.
NB-IoT Access	FAJ 121 4657	MI	Related	NB-IoT coexistence with standalone deployment mode is not supported. NB-IoT can be configured in one sector of a two-sectors combined cell.[1]

Feature Name	Feature Identity	Access Type	Relationship	Description
OTDOA Control Plane Location Support	FAJ 121 1793	LTE	Related	The OTDOA Control Plane Location Support feature requires `SectorCarrier.prsEnabled` to be configured as described in Parameters for Combined Cell.
OTDOA User Plane Location Support	FAJ 121 0862	LTE	Related	The OTDOA User Plane Location Support feature requires `SectorCarrier.prsEnabled` to be configured as described in Parameters for Combined Cell
Quad Antenna Uplink Performance Package	FAJ 121 1744	LTE	Related	The Quad Antenna Uplink Performance Package feature is needed when the Combined Cell feature is used with four TX and four RX antennas.
Uplink Carrier Aggregation	FAJ 121 4425	LTE	Related	A combined cell with more than two sectors cannot function as an Uplink Carrier Aggregation SCell for a UE.
Uplink Interference Reporting	FAJ 121 4157	LTE	Related	Uplink Interference Reporting does not provide observability for antenna branches on sector carriers in combined cell with some or all of its antennas disabled.
Atmospheric Duct Interference Reduction	FAJ 121 5065	LTE	Conflicting	Enabling both Atmospheric Duct Interference Reduction and Combined Cell is not supported.
Cross-Sector Antenna Sharing Redundancy	FAJ 121 1040	LTE	Conflicting	The Cross-Sector Antenna Sharing Redundancy feature cannot be used in conjunction with the Combined Cell feature.
Dual Layer Beamforming Performance Package	FAJ 121 3000	LTE	Conflicting	TDD only: The Combined Cell feature cannot be configured if the Dual Layer Beamforming Performance Package feature is enabled in the same eNodeB.
Intelligent Power Emission Control	FAJ 121 5224	LTE	Conflicting	The Intelligent Power Emission Control feature must not be activated together with features that enable other RAN types and modes of operation with multiple RANs.
LTE Modulation-Aware Power Control	FAJ 121 5577	LTE	Conflicting	If the sector carriers on the same combined cell have different values for the `SectorCarrier.ltePowerBackoffMode` attribute, the LTE Modulation-Aware Power Control feature might be disabled.
LTE PIM Avoidance	FAJ 121 5448	LTE	Conflicting	PIM avoidance is applied on the cell level, which results in the muting of all sector carriers in the cell.
Maximum Cell Range beyond 100 km	FAJ 121 5124	LTE	Conflicting	The Combined Cell feature can not be used together with the Maximum Cell Range beyond 100 km feature.
Micro Sleep Tx	FAJ 121 3089	LTE	Conflicting	Micro Sleep Tx feature is a basic feature for LTE and is turned on by default. Disabling this feature with MO attribute on sector carrier level is not supported if Combined Cell feature is used. One of the sectors alone cannot be disabled. The Micro Sleep Tx feature can be disabled only on cell level by setting the feature in all sectors to disabled and can be enabled on cell level by setting the feature in any sector to enabled.
Octal Antenna Uplink Performance Package	FAJ 121 2052	LTE	Conflicting	The two features Octal Antenna Uplink Performance Package and Combined Cell cannot be active simultaneously in the same eNodeB.
Return Loss Antenna Supervision	FAJ 121 4578	LTE	Conflicting	The Return Loss Antenna Supervision feature is not supported for the Baseband to Radio Gateway or LC-05 9xCPRI radio connection combination.
Single Layer Beamforming Performance	FAJ 121 3002	LTE	Conflicting	TDD only: The Combined Cell feature cannot be configured if the Single Layer Beamforming Performance Package feature is enabled in the same eNodeB.

Feature Name	Feature Identity	Access Type	Relationship	Description
Package				
Supplemental Downlink for Carrier Aggregation	FAJ 121 3068	LTE	Conflicting	Combined cells cannot be configured as downlink only SCells.
Supplemental Downlink with License-Assisted Access	FAJ 121 4573	LTE	Conflicting	The Combined Cell feature must be deactivated when the Supplemental Downlink with License-Assisted Access is activated. Combined cells cannot be configured as downlink only SCells.
Uplink Coordinated Multi-Point Reception	FAJ 121 3043	LTE	Conflicting	The same sector carrier cannot be part of Combined Cell and Uplink Coordinated Multi-Point Reception.

(1) Only valid for a Combined Cell with a maximum of 2TX/2RX.

Hardware

No specific hardware requirements.

Limitations

The radio configurations compatible with a combined cell are described in Radio Node Configurations.

Combined Cell can support 5 MHz, 10 MHz, 15 MHz, and 20 MHz bandwidth. 5 MHz bandwidth can be configured for 4.6 MHz or 4.8 MHz. Selected radios can be found in Radio Node Configurations.

Note: 9 MHz bandwidth cannot be used with Combined Cell.

All sector carriers within a combined cell must have the same number of antennas both in the downlink and uplink directions. The number of downlink and uplink antennas can differ within the same sector carrier. A sector carrier with some or all of its antennas disabled is degraded or disabled, respectively, and it is not included in the combined cell.

If some antennas belonging to a sector carrier that is included in a combined cell are disabled, the system initiates automatic cell lock and unlock to exclude the affected sector carrier from the combined cell. Cell lock and unlock results in a short-term cell outage.

If all antennas belonging to a sector carrier that is initially excluded from a combined cell are enabled, the system initiates automatic cell lock and unlock to add the affected sector carrier to the combined cell. Cell lock and unlock results in short-term cell outage.

A given sector carrier can only be connected to one combined cell, it is not a feasible configuration to connect it to multiple combined cells.

The delay difference between sector carriers within a combined cell must be less than 50 µs, which corresponds to a maximum distance of 10 km (pure fiber length). Otherwise, the cell is not enabled, and the `Resource Configuration Failure` alarm is raised. To cease the alarm and to enable the combined cell, the difference in fiber length must be less than 10 km. This also means that the values of the `RfBranch.dlTrafficDelay` and the `RfBranch.ulTrafficDelay` MOs cannot differ more than 50000 between the different sector carriers within the same cell.

To benefit from the macrodiversity on the PUSCH and PDSCH data channel, the sector carriers must not be placed more than 1500 m apart.

Heterogeneous network deployment can be configured in conjunction with the Combined Cell feature. However, possible unequal sector powers at different radio types are not taken into account in the downlink sector selection.

Downlink interference mitigation is not performed for the secondary cell when using the Carrier Aggregation feature together with the Combined Cell feature, since all sectors of the secondary cell are used for the downlink data transmission in the combined cell.

The following configurations are supported for Baseband radio node only:

- Combined cell with 9-12 sectors with 2TX/2RX antennas in each sector
- Combined cell with up to six sectors with 4TX/4RX antennas in each sector

NB-IoT coexistence with standalone deployment mode is not supported.

The Combined Cell feature with in-band or guard band deployment modes NB-IoT coexistence has the following limitations:

- NB-IoT coexistence with an LTE combined cell is only supported for NB-IoT in-band and guard band deployment

modes.

- The host LTE cell must be configured with two sector carriers with 2TX/ 2RX. No other antenna combinations are supported.
- By default, without manual configuration, the first sector carrier of the host LTE combined cell is used for the hosted NB-IoT cell. Manual configuration is available in one of the sector carriers of the host LTE combined cell.

If two Radio Gateway units are used, the `EUtranCellTDD` MO or `EUtranCellFDD` MO must have its `SectorCarrier` MOs allocated to Radio units that are connected to the same Radio Gateway. The `Inconsistent Configuration` alarm is raised if the Combined Cell is allocated to both Radio Gateways.

Network Requirements

Regarding the supported LTE combined cell configurations, see details in Features for Radio Configurations.

Related Information

NB-IoT Cell Coexistence with LTE Combined Cell

3 Feature Operation of Combined Cell

> The Combined Cell feature allows multiple sector carriers to belong to the same cell.

The sector carriers can either be located indoors or outdoors. However, all sector carriers that are present in the same cell are considered as one logical cell with the same PCI. Figure 1 provides an overview of the feature concept.

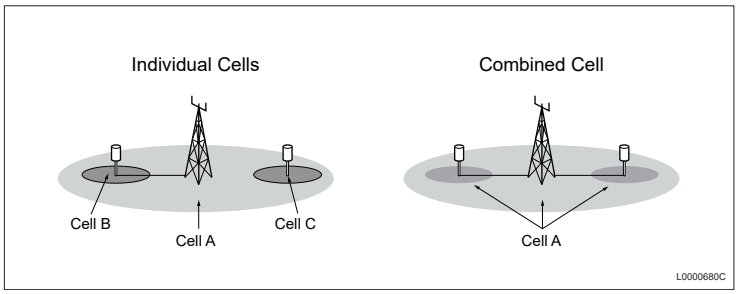

Figure 1 Overview Concept of the Combined Cell Feature

In a combined cell, multiple sector carriers can serve the same cell. The individual radios within a combined cell can have different output power setting configurations. In a combined cell, all radios are connected to the same Baseband Unit, however the different output power is not considered for the sector evaluation.

The Combined Cell feature supports uplink and downlink sector carrier selection within the same cell. In downlink, the feature supports selective muting of non-contributing sector carriers for the Physical Downlink Shared Channel (PDSCH), to reduce interference in neighboring cells. In uplink, if the UE is in an overlapping coverage area, the feature supports macrodiversity on the Physical Uplink Shared Channel (PUSCH) to improve the uplink transmissions.

The initial sector carrier selection for a UE is computed from the relative powers of the random access preambles received at the different sector carriers. After that, the received Physical Uplink Control Channel (PUCCH) Format 2 power measurements are used by the eNodeB to determine the transmission and reception sector carriers required for each UE.

3.1 Combination of Indoor and Outdoor Coverage

Figure 2 provides an example of the use of the Combined Cell feature to improve radio performance in a combination of indoor and outdoor environments. Multiple sector carriers of the same combined cell are used to provide coverage inside the building and macro coverage outside the building.

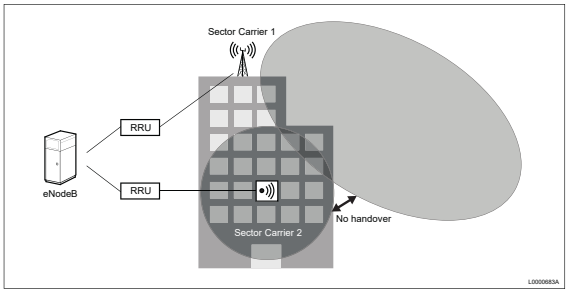

Figure 2 Example of Using Combined Cells in a Combination of Indoor and Outdoor Coverage Environments

The benefits of using the Combined Cell feature in this scenario are:

- Seamless mobility on entering and leaving the building.
- In coverage overlaps between sector carriers belonging to the same cell, an improved bit rate for a single UE can be expected compared to a single cell with a UE at cell border. The overlapping sector carriers provide a macrodiversity in this region.

3.2 Indoor Coverage

Figure 3 shows an example of the use of the Combined Cell feature to improve radio performance in an indoor environment. Multiple sector carriers of the same combined cell are used to provide indoor coverage to multiple floors of the same building.

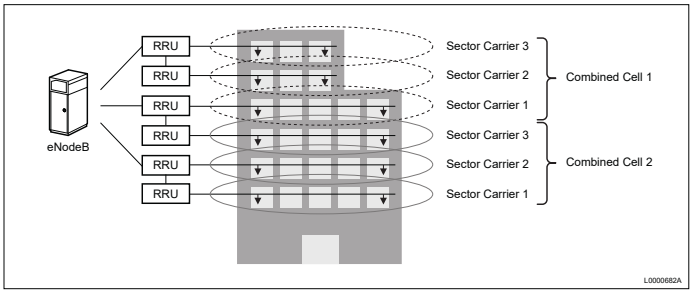

Figure 3 Example of Using Combined Cells in an Indoor Coverage Environment

The benefits of using the Combined Cell feature in this scenario are:

- Seamless mobility between the floors of the building covered by the same combined cell.
- In coverage overlaps between sector carriers belonging to the same cell, an improved bit rate for a single UE can be expected compared to a single cell with a UE at cell border. The overlapping sector carriers provide a macrodiversity in this region.

3.3 NB-IoT Cell Coexistence with LTE Combined Cell

> An in-band or guard band NB-IoT cell can exist in one sector of an LTE combined cell.

Only a combined cell with 2TX/2RX antenna configuration supports NB-IoT cell coexistence.

An LTE combined cell with two sector carriers can have an in-band or guard band NB-IoT cell configured. The NB-IoT cell only uses one of the sector carriers of the host LTE combined cell. The default value of `NbIotCell.sectorCarrierRef` is set to the first sector carrier of the host LTE combined cell. It is possible to manually set the `NbIotCell.sectorCarrierRef` to another cell. The attribute can only be set to one of the sector carriers of the host LTE combined cell.

3.3.1 Configure Sector Carrier of In-Band NB-IoT Cell

For the initial manual configuration, or the change of configuration of the sector carrier of the in-band NB-IoT cell:

Steps

1. Lock the in-band cell.
2. Lock the host cell.
3. Set the `NbIotCell.sectorCarrierRef` attribute to one of the elements of the `EUtranCellFDD.sectorCarrierRef` list of the host cell.
4. Unlock the host cell.
5. Unlock the in-band cell.

3.3.2 Configure Sector Carrier of Guard Band NB-IoT Cell

For the initial manual configuration, or the change of configuration of the sector carrier of the guard band NB-IoT cell:

Steps

1. Lock the guard band cell.
2. Lock the host cell.
3. Set the `NbIotCell.sectorCarrierRef` attribute to one of the elements of the `EUtranCellFDD.sectorCarrierRef` list of the host cell.
4. Unlock the host cell.
5. Unlock the guard band cell.

3.3.3 Validate Sector Carrier Configuration of Host LTE Cell

When the configuration of the `EUtranCellFDD.sectorCarrierRef` attribute changes, it must be validated that the sector carriers for the LTE cell contain the sector carrier for the NB-IoT in-band or guard band cell.

Steps

1. Set the `EUtranCellFDD.sectorCarrierRef` attribute.

   If the validation is successful, the `EUtranCellFDD.sectorCarrierRef` attribute is set successfully.

If validation fails, the following message is displayed: `At least one sector attached to NB-IoT cell <NbIotCell.cellId> is not attached to the host cell <EUtranCellFDD.cellId>!`

<NbIotCell.cellId> refers to the cell ID of the NB-IoT cell.

<EUtranCellFDD.cellId> refers to the cell ID of the host LTE cell.

Continue with the following steps:

2. Remove the `NbIotCell.sectorCarrierRef` attribute.
3. Set the `EUtranCellFDD.sectorCarrierRef` attribute again and repeat the procedure until validation is successful.

4 Network Impact of Combined Cell

> The feature improves coverage and mobility.

Capacity and Performance

The Combined Cell feature extends the indoor and outdoor coverage areas within the same cell. The cell capacity of the combined cell is not impacted. A combined cell has the same capacity as a cell that is not combined.

No neighbor relations need to be established between sector carriers of the same combined cell. This enables a reduction in the number of handovers.

Interfaces

No impact.

Other Network Elements

No impact.

5 Parameters for Combined Cell

> The feature introduces new and uses already existing MOM attributes.

Table 2 Feature Description

Parameter	Type	Description
ENodeBFunction.combCellSectorSelectThreshRx	Introduced	See MOM description. Sectors that are close to the UE contribute to the quality of the call if they are selected. Sectors that are selected but cannot hear the UE, do not affect the quality of the call, but contribute to unnecessary interference in downlink. At the sector selection evaluation, sectors that cannot hear the UE, have a power related to the noise floor. The best sector normally has a power similar to the PUCCH nominal power. The thresholds are used to select only the sectors that are contributing to the quality of the call.
ENodeBFunction.combCellSectorSelectThreshTx		
EUtranCellFDD.altitude	Affected	If all sector carriers in a combined cell have the same output power, for example in indoor scenario or 3-sector site scenario, it is assumed that the antennas of different sector carriers are located close to each other. Therefore, the position parameter is set using any of the antennas position. If there is one macro sector that has the highest power, the position parameter is set using the antenna position of the macro sector.
EUtranCellFDD.latitude		
EUtranCellFDD.longitude		
EUtranCellTDD.altitude		
EUtranCellTDD.latitude		
EUtranCellTDD.longitude		
EUtranCellFDD.sectorCarrierRef	Affected	Combined cell is configured by setting the `sectorCarrierRef` to more than one `SectorCarrier`.
EUtranCellTDD.sectorCarrierRef		
SectorCarrier.noOfRxAntennas	Affected	For the Combined Cell feature this parameter can be set to 0, 1, 2, or 4.
SectorCarrier.noOfTxAntennas		
SectorCarrier.prsEnabled	Affecting	For combined cells consisting of macro sector carriers and small sector carriers, set `SectorCarrier.prsEnabled` to `true` for the macro sector carrier only. The other sector carrier or sector carriers have `SectorCarrier.prsEnabled` = `false`. Transmission of PDSCH is still not scheduled on the same subframe in any of the sector carriers to prevent undesirable interferences.
EUtranCellFDD.commonSrPeriodicity	Affecting	Unlocking combined cells can be rejected because of PUCCH resource shortage. This results in a `Resource Configuration Failure` alarm. Required and available PUCCH resources for a combined cell depends on the following: – Number of combined cells per node, and the number of `SectorCarrier` configured per combined cell. – Cell bandwidth. – Setting of the following parameters: • `EUtranCellFDD.commonSrPeriodicity`
EUtranCellFDD.noOfPucchCqiUsers		
EUtranCellFDD.noOfPucchSrUsers		

Parameter	Type	Description
EUtranCellTDD.commonSrPeriodicity		• EUtranCellTDD.commonSrPeriodicity • EUtranCellFDD.noOfPucchCqiUsers • EUtranCellTDD.noOfPucchCqiUsers • EUtranCellFDD.noOfPucchSrUsers • EUtranCellTDD.noOfPucchSrUsers It is recommended that the EUtranCellFDD.noOfPucchCqiUsers and the EUtranCellTDD.noOfPucchCqiUsers value are not lower than the EUtranCellFDD.noOfPucchSrUsers and the EUtranCellTDD.noOfPucchSrUsers value. For more information about dependencies between the parameters, see Control Channel Dimensioning.
EUtranCellTDD.noOfPucchCqiUsers		
EUtranCellTDD.noOfPucchSrUsers		

6 Performance of Combined Cell

> The feature is associated with performance management entities measuring handover.

KPIs

This feature has no associated Key Performance Indicators.

Counters

- `pmActiveUeDlSectorCarrierDistr`
- `pmActiveUeDlSumSectorCarrier`
- `pmActiveUeUlSectorCarrierDistr`
- `pmActiveUeUlSumSectorCarrier`

Events

This feature has no associated events.

7 Activate Combined Cell

> To use the feature, it must be activated following the usual feature activation procedure.

Prerequisites

- The license key is installed in the node.
- CCTR is active for at least one week before this procedure to collect enough troubleshooting data.

Steps

1. Set the `FeatureState.featureState` attribute to `ACTIVATED` in the `FeatureState=CXC4011368` MO instance.

After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

8 Deactivate Combined Cell

> If the feature is no longer needed, it can be deactivated following the usual feature deactivation procedure. It must also be deactivated before the activation of any conflicting feature.

Prerequisites

CCTR is active for at least one week before this procedure to collect enough troubleshooting data.

Steps

1. Set the `FeatureState.featureState` attribute to `DEACTIVATED` in the `FeatureState=CXC4011368` MO instance.

After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

9 Engineering Guidelines for Combined Cell

> Engineering guidelines for Combined Cell contain hardware configuration examples and feature optimization.

9.1 Feature Optimization

Power measurements on PUCCH Format 2 containing CQI reports are needed to perform sector selection. Therefore, it is recommended that the value of `EUtranCellFDD.noOfPucchCqiUsers` and `EUtranCellTDD.noOfPucchCqiUsers` are set to no less than `EUtranCellFDD.noOfPucchSrUsers` and `EUtranCellTDD.noOfPucchSrUsers` to fully benefit from the Combined Cell feature.

For those UEs that cannot be configured with PUCCH Format 2 resources, that is PUCCH CQI, the sector selection is only performed at Physical Random Access Channel (PRACH). These UEs are not updated until the next random access procedure, due to the lack of PUCCH Format 2 power measurements.

9.2 Hardware Configurations

A given sector carrier can only be connected to one combined cell, it is not a possible configuration to connect it to multiple Combined Cells.

Examples of hardware configurations using the Combined Cell feature are provided in the following illustrations:

- Figure 4
- Figure 5
- Figure 6
- Figure 7
- Figure 8
- Figure 9
- Figure 10

Figure 4 provides an example configuration of one combined cell with eight sector carriers, 2TX/2RX, cascaded radios.

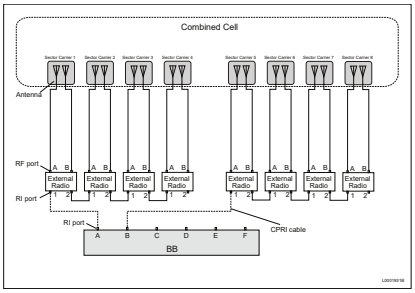

Figure 4 One Combined Cell with Eight Sector Carriers, 2TX/2RX, Cascaded Radios

Figure 5 provides an example configuration of one combined cell with eight sector carriers, 2TX/2RX, mixed configuration cascaded radios.

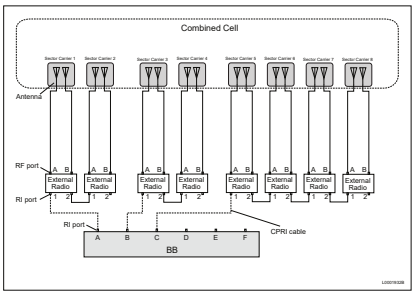

Figure 5 One Combined Cell with Eight Sector Carriers, 2TX/2RX, Mixed Configuration Cascaded Radios

Figure 6 shows an example configuration between two combined cells, one with three sector carriers 2TX/RX, and one with four sectors 4TX/4RX, with cascaded radios.

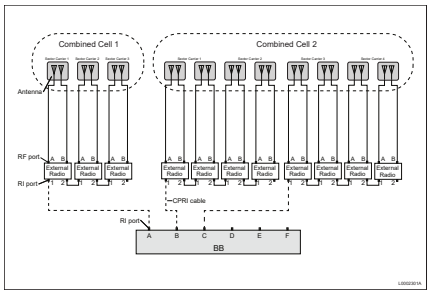

Figure 6 Two Combined Cells with Different Configurations

Figure 7 shows an example configuration of one combined cell with 12 sector carriers, 2TX/2RX, cascaded radios.

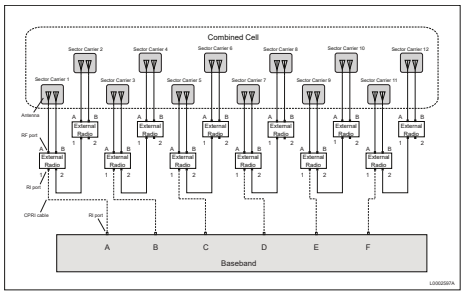

Figure 7 One Combined Cell with 12 Sector Carriers, 2TX/2RX, Cascaded Radios

Figure 8 shows an example configuration of one combined cell with six sector carriers, 4TX/4RX.

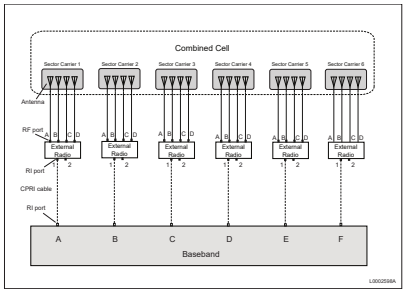

Figure 8 One Combined Cell with Six Sector Carriers, 4TX/4RX

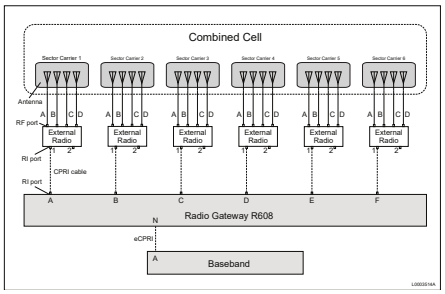

Figure 9 One Radio Gateway and a Combined Cell with Six Sector Carriers

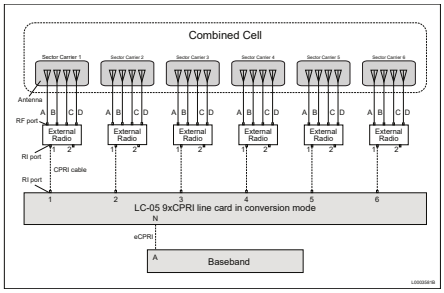

Figure 10 One LC-05 9xCPRI Line Card in Conversion Mode and a Combined Cell with Six Sector Carriers

An EUtranCell which has multiple sector carriers must have all sector carriers on RUs that are connected to the same Radio Gateway or LC-05 9xCPRI line card in conversion mode.

For more information about supported hardware configurations using the Combined Cell feature, see Features for Radio Configurations.

Appendix A: Feature Change History

> This section lists changes that affected this feature and the impact it had on the network.

Appendix A.a: 20.Q2: Combined Cell with 24 Antennas

> The enhancement enables combined cell support for up to 12 sectors and 24 antennas on Baseband Radio Nodes.

Feature Name:	Combined Cell
Feature Identity:	FAJ 121 3025
Value Package Name:	Combined Cell
Value Package Identity:	FAJ 801 0407
Node Type:	Baseband Radio Node
Access Type:	LTE

Benefits

This feature enhancement provides support for 24 antenna branches and for the following sector configurations:

- Up to 12 sectors with two RX and two TX antennas
- Up to six sectors with four RX and four TX antennas

The main benefits of this enhancement are the extended flexibility of combined cell configuration, and the reduced cell planning.

Capacity and Performance

Cell coverage is extended because of the greater number of sectors.
The number of inter-cell handovers decreases because of the extended coverage.
Cell capacity is unaffected, but because more sectors share the same capacity, the mean sector capacity decreases.

Operation

No impact.

Interfaces

No impact.

Hardware

No special hardware requirements.

Other Network Elements

No impact.

Appendix A.b: 19.Q2: Coexistence of NB-IoT and Combined Cell

> This enhancement supports the use of NB-IoT in one dedicated cell that is involved in LTE Combined Cell deployment.

Feature Name:	Combined Cell NB-IoT Access
Feature Identity:	FAJ 121 3025 FAJ 121 4657
Value Package Name:	Combined Cell M-IoT RAN on LTE Base Package

Value Package Identity:	FAJ 801 0407 FAJ 801 1000
Node Type:	Baseband Radio Node
Access Type:	LTE MI

Benefits

An NB-IoT cell can be configured in in-band or guard band deployment mode for an LTE combined cell and carry the same benefits as an LTE combined cell. Only a combined cell with 2TX/2RX antenna configuration supports NB-IoT cell coexistence. The NB-IoT cell only uses one of the sector carriers of the host LTE combined cell.

Capacity and Performance

No impact.

Operation

The sector carrier used in the NB-IoT cell is configured statically by MOM.

The `NbIotCell.sectorCarrierRef` attribute is updated. It is applicable to all deployments.

Interfaces

No impact.

Hardware

No special hardware requirements.

Other Network Elements

No impact.

Appendix A.c: 19.Q1: Combined Cell with Extended Configuration

> The enhancement enables Combined Cell at Baseband sites with four Tx and four Rx antennas, with a maximum of four sectors.

Feature Name:	Combined Cell
Feature Identity:	FAJ 121 3025
Value Package Name:	Combined Cell
Value Package Identity:	FAJ 801 0407
Node Type:	Baseband Radio Node
Access Type:	LTE

Up to four Rx and four Tx antennas can be set up for each sector in a combined cell. The extended configuration is supported in both FDD and TDD modes.

The improvement enables Combined Cell to be deployed in a wider range of site configurations.

Capacity and Performance

No impact.

Operation

No impact.

Interfaces

No impact.

Hardware

No special hardware requirements.

Other Network Elements

No impact.

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